

ASSIGNMENT 16

ROLL NO:CSB24083

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```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];

int in = 0;
int out = 0;

sem_t empty;
sem_t full;

pthread_mutex_t mutex;

void *producer(void *arg)
{
    int item;

    for(int i = 1; i <= 10; i++)
    {
        item = i;
```

```
    sem_wait(&empty);

    pthread_mutex_lock(&mutex);

    buffer[in] = item;

    printf("Producer produced item %d at index %d\n", item,
in);

    in = (in + 1) % BUFFER_SIZE;

    pthread_mutex_unlock(&mutex);

    sem_post(&full);

    sleep(1);
}

pthread_exit(NULL);
}

void *consumer(void *arg)
{
    int item;

    for(int i = 1; i <= 10; i++)
    {

        sem_wait(&full);
```

```
        pthread_mutex_lock(&mutex);

        item = buffer[out];

        printf("Consumer consumed item %d from index %d\n", item,
out);

        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);

        sem_post(&empty);

        sleep(2);
    }

    pthread_exit(NULL);
}

int main()
{
    pthread_t producerThread;
    pthread_t consumerThread;

    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);

    pthread_mutex_init(&mutex, NULL);

    pthread_create(&producerThread, NULL, producer, NULL);
```

```
pthread_create(&consumerThread, NULL, consumer, NULL);
```

```
pthread_join(producerThread, NULL);
```

```
pthread_join(consumerThread, NULL);
```

```
sem_destroy(&empty);
```

```
sem_destroy(&full);
```

```
pthread_mutex_destroy(&mutex);
```

```
return 0;
```

```
}
```

OUTPUT

```
PS F:\AP projects> gcc -g -pthread 2multithread.c -o 2multithread
PS F:\AP projects> .\2multithread.exe
Producer produced item 1 at index 0
Consumer consumed item 1 from index 0
Producer produced item 2 at index 1
Consumer consumed item 2 from index 1
Producer produced item 3 at index 2
Producer produced item 4 at index 3
Consumer consumed item 3 from index 2
Producer produced item 5 at index 4
Producer produced item 6 at index 0
Consumer consumed item 4 from index 3
Producer produced item 7 at index 1
Producer produced item 8 at index 2
Consumer consumed item 5 from index 4
Producer produced item 9 at index 3
Producer produced item 10 at index 4
Consumer consumed item 6 from index 0
Consumer consumed item 7 from index 1
Consumer consumed item 8 from index 2
Consumer consumed item 9 from index 3
Consumer consumed item 10 from index 4
```